

Sachin Avutu

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Education

Drexel University

Bachelor of Science in Computer Science, Minor in Linguistics

Sep 2024 – Expected Jun 2029

Experience

Full-stack Developer

The Triangle - Drexel University Student Newspaper

Philadelphia, PA

May 2025 – Present

- Leading migration from WordPress to a custom CMS and database architecture, enabling scalable publishing and full platform ownership
- Built a headless CMS in Go with a React frontend and REST API, replacing WordPress and enabling scalable content management
- Constructed a Python ETL pipeline to migrate 11+ years of WordPress data (18k+ articles, 5k+ authors) into MariaDB, preserving data continuity despite upstream failures
- Developed a MinHash-based entity resolution system using NumPy and Numba, reducing duplicate records and improving dataset integrity in production

Software Engineer Co-op

eMoney Advisor

Radnor, PA

Sep 2025 – Mar 2026

- Delivered production features and resolved technical debt in a 1.3M+ LOC .NET codebase supporting integrations with 12,000+ financial institutions
- Partnered with senior engineers to develop and deploy eMoney's Rocket Mortgage integration into production
- Implemented NUnit-based test suites to catch regressions and stabilize production release, while following TDD practices
- Traced and corrected a partner API change that triggered a spike in client-facing errors using Elasticsearch and Kibana
- Orchestrated production deployments through Jenkins and Octopus within CI/CD pipelines

Projects

VectorDB

github.com/ssavutu/littlebrother [↗](#)

- Built an HNSW-based Vector Database from scratch, enabling sub-X ms vector search across N+ embeddings
- Achieved Y% recall vs brute-force cosine similarity while delivering Z× faster query performance on large datasets
- Tuned parameters (M, efConstruction, efSearch) to optimize latency-accuracy-memory tradeoffs under varying workloads
- Engineered memory and cache-efficient graph structures to minimize traversal overhead and improve query performance at scale

Minhashlib

github.com/ssavutu/minhashlib [↗](#)

- Released an MIT-licensed high-performance Minhash similarity detection Python library for use on large sets of strings
- Developed a high-performance MinHash library achieving 6x speedup over datasketch on CPU benchmarks while maintaining comparable accuracy
- Accelerated signature generation through vectorization and JIT compilation using Numba
- Integrated Minhashlib in a production-critical ETL pipeline at The Triangle

DragonFuel

github.com/ssavutu/DragonFuel [↗](#)

- Created a multipage nutrition tracking app using the Flutter UI framework
- Assembled a Python scraping pipeline using Selenium and BeautifulSoup to aggregate nutritional data
- Designed a local data layer using SQLite and JSON to support persistent, real-time application state

RxBuddy

github.com/ssavutu/RxBuddy [↗](#)

- Prototyped a Flutter based medication reminder app MVP within a 24-hour hackathon
- Exposed a RESTful API via Flask enabling real-time image and JSON data exchange
- Leveraged Gemini API to extract and interpret text from prescription labels

Skills

Languages: Python, C#, C++, C, JavaScript, TypeScript, Dart, SQL, HTML, CSS

Technologies: .NET, NUnit, MariaDB, NumPy, Numba, Flask, Flutter, React, Selenium, BeautifulSoup, SQLite

Tools: Linux, Git, GitHub, Bitbucket, Jira, Confluence, Octopus, Jenkins, Elasticsearch